

Smart Engagement Analyzer

Student Name: Aastha Nagar

Registration No.: 23BAI11205

Course: Computer Vision (CSE3010)

Date: 26 March 2026

Project Repository: <https://github.com/AasthaNagar23/Smart-Engagement-Analyzer>

1. Project Overview

The Smart Engagement Analyzer is a Computer Vision–based real-time system that analyzes user engagement using webcam video input. The system captures live video frames and processes facial features to determine the user’s attention level.

The project focuses on detecting facial behavior such as eye movement, head orientation, and facial landmarks to classify engagement states like attentive, distracted, or drowsy.

This system demonstrates practical applications of Computer Vision in human behavior analysis and intelligent monitoring systems.

2. Objectives

- To capture real-time video using a webcam
- To detect and analyze human facial features using Computer Vision
- To track eye movement and head orientation
- To estimate user engagement level based on facial cues
- To apply image processing techniques for frame-wise analysis

3. Tools and Technologies

- **Programming Language:** Python 3.10+
- **Libraries:**
 - OpenCV – real-time image and video processing
 - MediaPipe – facial landmark detection
 - NumPy – numerical operations
- **IDE:** Visual Studio Code
- **Version Control:** Git & GitHub

4. Methodology

Step 1: Video Capture

A webcam is used to capture real-time video frames.

Step 2: Frame Processing

Each frame is processed individually using OpenCV for analysis.

Step 3: Face Detection

Facial landmarks are detected using MediaPipe Face Mesh, identifying key points such as eyes, nose, and face contour.

Step 4: Feature Extraction

Important features extracted include:

- Eye aspect ratio (for blink/attention detection)
- Head position and orientation
- Facial landmark movement patterns

Step 5: Engagement Analysis

Based on extracted features, the system classifies engagement into:

- Attentive
- Distracted
- Drowsy

Step 6: Output Visualization

Results are displayed in real-time on the video feed.

5. Computer Vision Concepts Used

This project directly applies the following Computer Vision concepts:

- Image Acquisition (webcam input)
- Face Detection and Landmark Detection
- Feature Extraction from images
- Real-time video frame processing
- Motion analysis of facial features
- Object detection (face as object)

6. Features

- Real-time webcam-based processing
- Face and eye tracking
- Engagement level detection
- Lightweight and fast processing
- Uses deep learning–based landmark detection (MediaPipe)

7. Applications

- Online learning engagement monitoring
- Driver drowsiness detection systems
- Workplace attention tracking
- Human-computer interaction systems
- Smart surveillance systems

8. Conclusion

The Smart Engagement Analyzer demonstrates how Computer Vision techniques can be used to interpret human facial behavior from real-time video input. By analyzing facial landmarks and motion patterns, the system effectively estimates user engagement levels.

This project successfully applies core Computer Vision concepts such as face detection, feature extraction, and real-time image processing, making it relevant to modern intelligent monitoring systems.